Classwork 07//KL Divergence in DDPM

Student Name: ______________________________ **Points:** 30

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KL Divergence for Denoising Diffusion Model (DDPM) is given by

$$\mathrm{KL}(q,p) = \underbrace{\mathbb{E}_{q(\mathbf{x}_0,\mathbf{x}_1,\cdots,\mathbf{x}_T)}\Big[-\log p(\mathbf{x}_T) - \sum_{t=1}^{T}\log\frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1})}\Big]}_{=L} + \text{constant}$$

Show that L can be written as

$$\mathbb{E}_{q(\mathbf{x}_0,\mathbf{x}_1,\cdots,\mathbf{x}_T)}\Big[KL(q(\mathbf{x}_T|\mathbf{x}_0)||p(\mathbf{x}_T)) + \sum_{t>1} KL(q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)||p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)) - \log p_\theta(\mathbf{x}_0|\mathbf{x}_1)\Big]$$

Use the fundamental definition of KL divergence for proving.

Proof:

The logarithm identity can be used for proving. First, we note that $\log\frac{a}{b} = \log a - \log b$. Use $\mathbf{x}_0, \mathbf{x}_1, \cdots, \mathbf{x}_T = \mathbf{x}_0 : \mathbf{x}_T$ notation.
We also note that $KL(q,p) = \mathbb{E}_q[\log\frac{q}{p}]$.
Then,

$$\begin{aligned} L &= \mathbb{E}_{q(\mathbf{x}_0:\mathbf{x}_T)}\Big[-\log p(\mathbf{x}_T) - \sum_{t=1}^{T}\log\frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1})}\Big] \\ &= \mathbb{E}_{q(\mathbf{x}_0:\mathbf{x}_T)}\Big[-\log p(\mathbf{x}_T) - \sum_{t=1}^{T}\log p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) + \sum_{t=1}^{T}\log q(\mathbf{x}_t|\mathbf{x}_{t-1})\Big] \end{aligned}$$

Also, we know from Bayes' theorem that

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)q(\mathbf{x}_t|\mathbf{x}_0)}{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}$$

Thus,

$$\sum_{t=}^{T} \log q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \log q(\mathbf{x}_1|\mathbf{x}_0) + \sum_{t=2}^{T}\left[\log q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) + \log q(\mathbf{x}_t|\mathbf{x}_0) - \log q(\mathbf{x}_{t-1}|\mathbf{x}_0)\right]$$

If we expand the sum, then consecutive $\log q(\mathbf{x}_t|\mathbf{x}_0)$ with $+$ and $-$ will cancel out.
Hence,

$$\sum_{t=}^{T} \log q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \sum_{t=2}^{T} \log q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) + \log q(\mathbf{x}_T|\mathbf{x}_0)$$

Then

$$L = \mathbb{E}_q(\mathbf{x}_0 : \mathbf{x}_T)\left[-\log p(\mathbf{x}_T) + \sum_{t=2}^{T} \log q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) + \log q(\mathbf{x}_T|\mathbf{x}_0) - \sum_{t=1}^{T} \log p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)\right]$$

Split the sum over p_θ, we have

$$-\sum_{t=1}^{T} \log p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = -\log p_\theta(\mathbf{x}_0|\mathbf{x}_1) - \sum_{t=2}^{T} \log p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$$

Thus

$$\begin{aligned}
L = \mathbb{E}_q(\mathbf{x}_0 : \mathbf{x}_T)\Bigg[&-\log p(\mathbf{x}_T) + \sum_{t=2}^{T} \log q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) + \log q(\mathbf{x}_T|\mathbf{x}_0) \\
&- \log p_\theta(\mathbf{x}_0|\mathbf{x}_1) - \sum_{t=2}^{T} \log p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)\Bigg] \\
L = \mathbb{E}_q(\mathbf{x}_0 : \mathbf{x}_T)&\left[\frac{\log q(\mathbf{x}_T|\mathbf{x}_0)}{\log p(\mathbf{x}_T)} + \sum_{t=2}^{T} \log \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} - \log p_\theta(\mathbf{x}_0|\mathbf{x}_1)\right]
\end{aligned}$$

Each log is argument to KL as per its definition, thus, taking expectations over $q(\mathbf{x}_0 : \mathbf{x}_T)$

$$L = \mathbb{E}_q(\mathbf{x}_0 : \mathbf{x}_T)\left[KL(q(\mathbf{x}_T|\mathbf{x}_0)||p(\mathbf{x}_T)) + \sum_{t=2}^{T} KL(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)||p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)) - \log p_\theta(\mathbf{x}_0|\mathbf{x}_1)\right]$$